

Beyond Backtest Illusions: Executable Research Integrity Guards and Counterfactual Auditing for Financial AI Agents

Howard Ye^{1*} Shwai He^{2*}

^{1,2}Open Quantitative Integrity Initiative (`fin-skills` & Multimodal Financial NLP Group)
{howard, shwaihe}@fin-skills.org (*Equal contribution)

Abstract

Large language model (LLM) agents increasingly automate quantitative financial research, yet reported backtest Sharpe ratios routinely collapse out-of-sample. We identify two distinct failure modes driving this collapse: **Type I Mechanical Shortcuts** (look-ahead joins, unadjusted corporate actions, survivorship bias, and pre-training window contamination) and **Type II Empirical & Microstructure Illusions** (trusting retail echo-chambers where high-reach influencers carry negative alpha, and ignoring market execution rules such as $T + 1$ lock-ups and board-lot granularity). To solve both, we introduce **fin-skills**, an open-source library of **129 source-verified skills**, **36 executable guards** behind a unified `Bundle.check()` contract, and **53 MCP agent tools**, paired with an external **Counterfactual Audit Protocol** that evaluates agent submissions without using the library to grade itself. Across synthetic worlds with 12 planted defects (honest ceiling Sharpe 1.63 vs. wrong-side join 71.5), our guards achieve **100% recall (36/36) with zero false alarms**. Crucially, session-by-session clock replay exposes leaky pipelines that score Sharpe 22.80 on held-out static years but lose -1.17 bps/day (Sharpe -0.58) live. In a controlled 12-run multi-model pilot, `fin-skills` reduces strong-model (Claude Opus) Sharpe reporting error by **84%** ($0.19 \rightarrow 0.03$), but uncovers a striking **Compliance Paradox**: equipping weak models (Claude Haiku) with complex documentation alone yields the worst distortion (11.95 Sharpe gap) as agents *hallucinate citations to guards they never executed*. Finally, we validate external domain utility on **1.72M bilingual social posts** and **1,445 verified A-share KOL track**

records, demonstrating a $3.96\times$ follower paradox that drives unweighted NLP into negative correlation (Rank IC = -0.0318 , $p = 1.74 \times 10^{-20}$) until reversed by Bayesian credibility gating (Rank IC = $+0.0104$, $p = 0.001$).

1 Introduction

Autonomous LLM agents equipped with code interpreters and financial data feeds can formulate hypotheses, engineer features, and report backtest performance in minutes. However, high reported Sharpe ratios in computational finance rarely survive live deployment [2, 1, 3]. In this paper, we demonstrate that AI research agents fail through two orthogonal mechanisms:

- 1. Type I: Mechanical Research Shortcuts.** Agents silently exploit structural artifacts in tabular market workspaces: joining post-close news commentary to the same session’s open-to-close return, training on pre-computed LLM scores whose fitting window overlaps the evaluation period, filtering universes by future survival, or computing factors on unadjusted split prices.
- 2. Type II: Empirical & Microstructure Illusions.** Even when timestamps are strictly causal, agents misinterpret noisy social streams and market rules. On financial social platforms, viral retail sentiment peaks at local price tops, while real execution is constrained by $T + 1$ settlement freezes, 100-share board lots, asymmetric stamp duties, and secondary-market ETF premium bubbles.

Inspired by behavioral testing in NLP [4] and open-source ML systems design [5], we present **fin-skills** alongside an external **Counterfactual Microstructure Audit Methodology**. Rather than claiming to manufacture market alpha, our framework enforces verifiable research integrity. Our four contributions are:

- **The fin-skills Architecture (§2):** 129 verified skills, 36 executable guards (`Bundle.check()`), and 53 JSON/MCP tools validated by 3,082 unit tests across Python 3.10–3.13.
- **External Counterfactual Auditing (§3):** Four black-box perturbation operators and session-by-session clock replay that detect look-ahead and same-session leakage without inspecting agent code.
- **The Compliance Paradox (§3.1):** Controlled multi-model experiments showing that while strong models benefit in reporting precision, weak models given un-gated documentation hallucinate compliance with guards they never ran.
- **Real-World Domain Validation (§4):** Reproducible empirical audits across 1.72M bilingual posts (1,445 KOLs) and A-share microstructure execution guards.

2 The fin-skills Architecture

`fin-skills` organizes quantitative domain knowledge and executable verification into three layers across seven domain packages (`fin-core`, `fin-ml`, `fin-market-data`, `fin-alt-data`, `fin-china`, `fin-strategies`, and `fin-llm`):

1. Source-Verified Skills (129 Specifications). Every skill conforms to a strict 6-field schema carrying explicit verification dates (`verified_on`) and authoritative primary sources. Skills document silent failure modes in standard libraries (e.g., day-count conventions, fractional differentiation truncation [3], and point-in-time SEC Form 4 / 13F acceptance timestamps).

2. Executable Guards (`Bundle.check()`).

To bridge passive reading and active verification, `fin-skills` provides 36 executable Python guards registered under a uniform slot-binding contract:

$$\mathcal{R} = \text{Bundle}(\mathbf{r}, \mathbf{P}, \tau, \dots).check() \in \{\text{PASS}, \text{WARN}, \text{FAIL}\}^{36} \quad (1)$$

Guards inspect typed slot inputs ($\mathbf{r}$ returns, $\mathbf{P}$ OHLCV bars, τ turnover, corporate actions, and point-in-time join keys) and return structured diagnostics.

3. MCP Tools Point-in-Time Collectors.

All 36 guards, portfolio/econometric algorithms, and public disclosure collectors (`fin_skills.collect` supporting RSS, static HTML, SEC EDGAR, and House PTR filings with SQLite revision tracking) export native Model Context Protocol (MCP) schemas (53 tools total).

3 Counterfactual Audit Methodology & The Compliance Paradox

A fundamental methodological requirement is that *an evaluation harness must not grade agent submissions using the same library guards provided to the agent*. We therefore construct an independent ground-truth **Seeded Market Generator** (`benchmarks/agent_study/`) spanning 60 equities over 2017–2023 with four planted high-reward shortcuts: (i) a news feed stamped with the session it describes, (ii) an LLM sentiment score trained through 2020-12-31, (iii) a listing table containing future delisting dates, and (iv) unadjusted split prices.

Calibrating the Honest Ceiling vs. Shortcuts.

Because the data-generating process is known exactly, we measure the true theoretical ceiling: trading the generator’s latent drift with a strict 1-day lag earns **Net Sharpe** 1.63. By contrast, joining same-session news earns **Net Sharpe** 71.5. Any reported Sharpe far above 1.63 is mathematical proof of a shortcut.

Reference Pipeline	Leakage	Paper OOS	Sharpe Clock Replay
Honest (lagged feed, split-adj.)	0.00	1.63	12-run agent pilot across seeds {11, 23, 37}:
Leaky (same-day join, full scaler)	0.78	22.80*	+5.07 bps/d (SR 2.99)

Table 1: Instrument calibration (*illusory paper Sharpe). Holding out a future year fails to punish same-session leakage (22.80); only physical clock replay exposes it (−0.58).

Experimental Arm ($n = 3/\text{cell}$)	$ \Delta\text{Sharpe} $	Leakage	Same-Day Contam. LLM	mean $ \Delta\text{Sharpe} $
Claude Opus (No Library)	0.19	0.00	0.00	(11.95 $\pm$ 0.43)
Claude Opus + <code>fin-skills</code>	0.03	0.00	0.00	0.24 while earning 27.96 in one run, and
Claude Haiku (No Library)	0.57	0.00	0.00	claiming 2/3 while losing −4.86 in another.
Claude Haiku + <code>fin-skills</code>	11.95	0.00	0.42	Crucially, Haiku cited guards in its

Table 2: 12-run agent pilot across seeds {11, 23, 37}. $|\Delta\text{Sharpe}| = |\text{Claimed} - \text{Recomputed}|$. Weak model + full library yields the worst distortion (11.95) due to hallucinated compliance.

Four Counterfactual Perturbations. Given a submitted function $\mathbf{W} = f(\mathcal{D})$, our oracle (`oracle.py`) measures four black-box invariants:

1. **Future Redraw (leakage_rate):** Redraw market paths after cut date t_c ; measure fraction of pre-cut weights $\mathbf{W}_{t \leq t_c}$ that change across 3 redraws.
2. **Single-Session Redraw (same_session_rate):** Redraw session t alone and inspect $\mathbf{W}_t$. This catches same-day post-close joins invisible to cut-date tests.
3. **Input Scramble (uses_⟨file⟩):** Permute individual input files to establish causal dependence on contaminated features (e.g., `llm_score.csv`).
4. **Session-by-Session Clock Replay:** Withhold future sessions physically by feeding data day-by-day (`live_replay.py`).

3.1 Pilot Findings & The Compliance Paradox

Table 1 proves that holding out a future year (2023) does not prevent same-session leakage (22.80 paper

Sharpe Clock Replay). Table 2 reports our 12-run agent pilot across seeds {11, 23, 37}:

• Strong Models Gain Reporting Precision:

Claude Opus avoided all four traps independently, and `fin-skills` reduced its Sharpe reporting discrepancy by **84%** (0.19 $\rightarrow$ 0.03).

• The Compliance Paradox: Equipping Claude Haiku with the full library produced the worst reporting distortion in the study

—claiming Sharpe while earning 27.96 in one run, and claiming 2/3 while losing −4.86 in another. Crucially, Haiku cited guards in its final report that it never executed. Passive prompt guidelines induce hallucinated compliance unless backed by mandatory runtime gates (`open_agent_runner.py`).

- **Realized Returns Do Not Differ:** Across 6 held-out 2023 live replays (40 sessions), the library yielded 1 win (+4.44% vs. +3.90%), 1 loss (−0.64% vs. +2.36%), and 1 tie (+0.30% vs. +0.55%). Integrity tools prevent false discoveries; they do not manufacture alpha.

4 Real-World Domain Validation: KOL Credibility & Microstructure

To prove that Type II empirical and microstructure illusions govern real markets, we audit **1.72M bilingual posts** (Chinese Xueqiu and US StockTwits) paired with **1,445 verified A-share KOL track records** (`benchmarks/real_world_kol_audit.py`).

The 3.96× Follower Asymmetry Paradox. Table 3 reveals why standard financial NLP fails on social media: accounts in `TIER_CONTRARIAN_INDICATOR` ($N = 98$, mean 5-day IC = −0.2130, win rate 28.49%) average **55,988.6 followers**—3.96× more than **TIER_1_CORE_ALPHA** researchers (14,121.9 followers, IC = +0.2158)! Consequently, unweighted Multimodal Attention Networks (MMAN) suffer catastrophic negative correlation

KOL Credibility Tier	Count	Mean Fans	5D Win%	5D Rank IC	Numerical Parity, Scaling & Conclusion		
TIER_1_CORE_ALPHA	70	14,121.9	68.47%	+0.1929	In benchmarks/parity_and_cost_bench.py), fin-skills achieves 8/8 exact numerical parity against scipy, statsmodels, and analytical closed forms (maximum discrepancies of 0.0 to 2.12×10^{-10} across SLSQP minimum-variance QP, parametric/historical VaR & ES, fractional differentiation weights, and Augmented Dickey-Fuller statistics). Across dataset sizes $N \in \{250, 1000, 5000, 25000\}$ rows, Bundle.check() exhibits sub-linear scaling $O(N^{0.2637})$, completing in 111.17 ms (3.66 MB peak RAM) at $N = 25,000$ rows (~ 100 years of daily bars).	Experiment	E3
TIER_2_RESEARCHER	115	31,994.4	70.92%	+0.1141			
TIER_0_ELITE_KOL	31	186,610.1	61.65%	+0.0008			
TIER_MEDIA_AGGREG.	11	107,354.3	46.98%	+0.0074			
TIER_NEUTRAL_RETAIL	1,120	71,251.4	46.72%	+0.2130			
TIER_CONTRARIAN	98	55,988.6	28.49%	+0.0140			

Table 3: Empirical stratification of 1,445 A-share KOLs. Contrarian indicators average $3.96\times$ more followers than Tier-1 Alpha researchers (55,989 vs. 14,122), explaining why unweighted social NLP loses money.

Model / Representation Variant	AUC	Acc.	5D Rank IC
MMAN Unweighted Fact Surge	0.4850	48.21%	-0.03184**
MMAN Baseline (Price + Raw Text)	0.5016	49.82%	+0.00609
MMAN + Bayesian Credibility Gate	0.5033	51.13%	+0.01040**
Chinese RoBERTa (Raw Social Text)	0.5075	50.67%	+0.00842
Chinese RoBERTa (LLM-Distilled)	0.5121	50.42%	+0.01862

Table 4: Out-of-sample predictive ablation. Unweighted fact extraction suffers severe negative Rank IC ($-0.03184, p = 1.74 \times 10^{-20}$); Bayesian credibility gating reverses it ($+0.04224$ net IC shift). LLM text distillation ($N = 31,505$) doubles RoBERTa Rank IC.

(Rank IC = $-0.03184, p = 1.74 \times 10^{-20}$, Table 4). Applying conjugate Beta-Binomial shrinkage and inverting statistically verified contrarians flips predictive power to Rank IC = $+0.01040$ ($p = 0.001$). Furthermore, LLM semantic distillation across 31,505 instances raises numerical fact density ($1.049 \rightarrow 1.078$ per 100 chars) and more than doubles Chinese RoBERTa Rank IC ($+0.00842 \rightarrow +0.01862$).

Institutional Microstructure Guard Enforcement. When deployed in an 80/20 Core-Satellite A-share portfolio (2023–2026 out-of-sample), enforcing six domain guards ($T+1$ lockups, 100-share board lots, asymmetric sell-side stamp duties, QDII secondary-market premium caps $\leq 1.5\%$, cash drag, and predictability stratification) achieves an annualized Sharpe of **1.34** and maximum drawdown of -6.88% (vs. CSI 300 benchmark -27.65%), scoring **95.0/100** on live paper-ledger audits.

First, as demonstrated in §3.1, fin-skills is an integrity verification toolkit rather than an alpha-generation engine: in held-out live replays where underlying strategies lack edge, guard enforcement does not increase realized returns. Second, our controlled agent pilot ($n = 12$ runs across two frontier model tiers) establishes clear measurement discrimination and uncovers hallucinated compliance, but full statistical power across open-weight model families requires completing our pre-registered 3-condition expansion (Experiment E2: 12 markets $\times$ 3 model sizes $\times$ 3 conditions $\times$ 3 repetitions, Appendix B). Third, real-world social sentiment dynamics evolve as platforms alter recommendation algorithms, requiring periodic re-estimation of Bayesian KOL priors.

Ethical Considerations & Broader Impact

All 1,445 social media author handles in our released benchmark fixture (kol_credibility_linguistic_fixture.json) are cryptographically pseudonymized via one-way SHA-256 digests (KOL_<10-hex>) with raw handles and stock-specific identifiers stripped to protect individual privacy. None of the tools, guards,

or empirical case studies in this paper constitute investment advice or automated live-money order execution; all broker-facing skills enforce strict paper-account sandboxing and human-in-the-loop kill switches. By exposing fabricated backtest Sharpe ratios and hallucinated audit citations, this work directly mitigates the risk of retail and institutional capital loss caused by deploying flawed AI-generated trading strategies.

References

- [1] David H Bailey, Jonathan Borwein, Marcos López de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69, 2017.
- [2] David H Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94–107, 2014.
- [3] Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.
- [4] Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. Beyond accuracy: Behavioral testing of NLP models with Checklist. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4902–4912, 2020.
- [5] Yue Zhao, Zain Nasrullah, and Zheng Li. PyOD: A Python toolbox for scalable outlier detection. *Journal of Machine Learning Research (JMLR MLOSS)*, 20(96):1–7, 2019.

A Experiment E1: Synthetic Defect Recall Across Random Seeds

In `benchmarks/guard_bench.py`, we evaluate guard recall across three independent synthetic market worlds generated with seeds $s \in \{17, 29, 43\}$, each containing 12 planted research defects spanning look-ahead joins, unadjusted corporate splits, survivorship filtering, non-monotonic as-of merges, cost implausibility, and regime look-ahead. Across all three worlds, `fin-skills` achieved **100% recall** (36/36 total defects detected), **0 false alarms** on clean control series, and **0 runtime exceptions** (`benchmarks/GUARD_ROBUSTNESS.json`).

B Experiment E2: Three-Condition Open-Weight Scaffold & Pre-Registered Hypotheses

To strictly isolate whether agent improvements stem from passive prompt documentation or active code execution, we implement `benchmarks/agent_study/open_agent_run` which evaluates any OpenAI-compatible open-weight server (vLLM/SGLang on Beacon) across three conditions:

- **Condition A (`no_library`):** Task prompt and raw market CSVs only.
- **Condition B (`skills_text_only`):** Injects relevant `SKILL.md` guidance into context while blocking executable guard tools.
- **Condition C (`skills_plus_guards`):** Provides both `SKILL.md` text and executable guard tools with immutable runtime provenance tracking to automatically flag hallucinated citations.

We freeze four pre-registered hypotheses prior to executing the full $12 \times 3 \times 3 \times 3$ sweep: (H1) shortcut incidence increases monotonically as model parameter count decreases; (H2) Condition B reduces reporting error for strong models but increases hallucinated guard citations for weak models; (H3) Condition C achieves the lowest $|\Delta \text{Sharpe}|$ across all model sizes; and (H4) realized live replay returns

show no statistically significant difference across Conditions A, B, and C.

C Experiment E3: Full Numerical Parity & Guard Runtime Scaling Tables

Table 5 details the exact numerical parity verification across all 8 financial and econometric primitives, alongside the full wall-clock timing and peak memory profile of `Bundle.check()` across dataset scales (`benchmarks/PARITY_AND_COST_RESULTS.json`).

Primitive	Reference Benchmark	Max $ \Delta $	Status
Equal Weight	Exact closed form ($1/N$)	0.00	PASS
Inverse Vol.	Exact sample σ_i^{-1}	0.00	PASS
Min-Var QP	<code>scipy SLSQP</code> convex QP	1.14×10^{-8}	PASS
Normal VaR/ES	<code>scipy.stats.norm</code> exact	6.94×10^{-18}	PASS
Hist. VaR/ES	Empirical quantile & tail mean	0.00	PASS
Hist. Vol.	<code>pandas</code> sample std ($ddof = 1$)	0.00	PASS
Frac. Diff. FFD	Closed-form binomial ($d = 0.4$)	1.11×10^{-16}	PASS
ADF Unit Root	<code>statsmodels.adfuller</code>	2.12×10^{-10}	PASS

Table 5: Complete E3 numerical parity verification (8/8 passed).